Blackwish asset generation research

Start here

Research checked 23 October 2026. A wide-scope reference for making, acquiring, transforming and validating creative assets across games, film, books, interactive media and physical-output handoffs.

The central finding: asset generation is not one operation and an asset is not one file. A useful pipeline preserves an editable source, records how derivatives were made, tests each derivative in its destination, and keeps rights and release authority separate from technical success. A beautiful preview can conceal an unusable rig, a missing font license, a broken audio loop or an impossible physical part.

Three reading routes

- **Choose a method:** asset map and production contract on pages 2-4; visual families on pages 5-12; procedural and AI methods on pages 13-14.
- **Prepare a usable result:** interchange on page 15; sound, voice and narrative on pages 16-18; destination constraints on pages 19-20; records and validation on pages 21-22.
- **Apply to Blackwish:** costs and worked examples on pages 23-24; three distinct Zen reviews on pages 25-27; the bounded Death Knell challenge on page 28; proposed next decisions on page 29.

Five rules worth retaining

1. Define the destination before choosing a generator. A browser icon, close-up film prop and printable model have different success conditions.
2. Preserve masters. Exporting a flat image, baked mesh or mixed audio file does not preserve all authoring structure.
3. Record transformations, dependencies and rights evidence. A filename or prompt is not a complete provenance record.
4. Validate at the level of the claim. A file parser cannot certify narrative quality, a visually pleasing render cannot certify geometry, and a license label cannot certify consent.
5. Keep authorization explicit. Generated does not mean validated, licensed, ratified or deployed.

How to read the evidence. Numbered source references link to the access ledger at the back. Documentation establishes a format or tool's described behavior, not independent performance or guaranteed interoperability. Original workflows, calculations, examples and proposed Blackwish fields are analytical recommendations, not measured production results. No current-price comparison or claim of literal exhaustiveness is made.

Authority boundary. This is research and hypothetical internal-consistency analysis only. It does not change Blackwish canon or live systems, run experiments on people, purchase anything, or authorize deployment. All examples are original, G-rated in intent and nonfantasy. The Zen and Death Knell sections are original analytical synopses, not quotations from a verified canonical artifact. Any later deployment remains subject to explicit manual ratification and its own applicable permissions.

Asset families and navigation

A production inventory should classify both **what the asset represents** and **what work remains**. A chair might be a concept sketch, rigged character prop, environment module, sound source, book illustration and printable miniature. These are related records, not interchangeable deliverables.

Family	Typical editable source	Key downstream question	Pages
2D and graphic design	Layers, paths, palettes, layout	Does it remain legible at actual size?	5-6
Geometry and surfaces	CAD or DCC scene, sculpt, material graph	Does shape and appearance survive export?	7-8
Characters and motion	Mesh, rig, weights, clips, captures	Does it deform and perform correctly?	9-10
Environments and capture	Modular kit, scene rules, photos or scans	Are scale, seams and coverage coherent?	11-12
Procedural and generative systems	Graphs, rules, inputs, model references	Can the result be traced and rebuilt?	13-14
Audio and voice	Recordings, sessions, stems, edit decisions	Is it clear, timed, licensed and intelligible?	16-17
Text and narrative	Structured text, branches, localization records	Is meaning consistent across contexts?	18
Shared production records	Dependencies, rights, tests, approvals	What exactly is this version allowed to do?	21-28

A second axis is how it was made

Handmade includes drawing, modeling, composing and writing. **Captured** includes photography, scanning, field recording and performance capture. **Procedural** means rules generate or assemble variation. **Generative AI** produces candidates through a learned model. **Acquired** means licensed or commissioned material. **Hybrid** is often the honest description: a scanned shape is remodeled, procedurally varied, hand-painted and exported by an automated build.

None of these labels is a quality rank. Handmade work can be inconsistent; a purchased asset can be expensive to adapt; procedural output can be repetitive; generated output can be difficult to revise precisely. Decide using the target, control requirements, provenance and total adaptation burden rather than the novelty of the method.

The rest of the reference

Production contract and gates 3-4. Define what success means and which evidence proves it. **Interchange 15.** Choose what crosses application boundaries. **Destinations 19-20.** Games, VR, film, books and manufacturing handoffs impose different constraints. **Economics and examples 23-24.** Compare retained value, not only generation speed. **Reviews 25-28.** Challenge assumptions before any commitment. **Decisions 29.** Keep the next move small and reversible. **Source ledger 30 onward.** Follow the exact source and its stated access limitation.

Define the asset contract before generation

An asset contract is a short specification of intended use, required representations and acceptance evidence. It prevents a visually impressive output from becoming an accidental promise to support every medium. This is a proposed working method, not an existing Blackwish implementation.

Describe the job

Start with a concrete sentence: "A movable, readable information sign for a small museum scene, with a matching print illustration and a spoken description." Then identify audiences, platforms, actual viewing distances, interaction, language coverage and accessibility requirements. Separate facts from preferences: the sign's text is factual content; its paint color may be a style choice; its physical dimensions may matter to a planned installation.

Specify what must remain editable

List what a future editor must change without rebuilding everything: text, colors, dimensions, facial expression, camera, voice line, timing, localization or material wear. If the generator cannot expose those controls, budget for reconstruction. A convincing image of a sign is not equivalent to editable typography; a flattened audio mix is not equivalent to separate dialogue and music stems.

Declare a representation set

A useful contract names an authoring master, a review preview, one or more interchange representations and each destination derivative. It also identifies nonvisual companions: collision, semantic labels, subtitles, accessibility descriptions, attribution, dependencies and tests. A preview should visibly identify its scope; do not let "approved appearance" become "approved in every destination."

Set measurable acceptance conditions

Prefer observable statements: no unresolved external dependencies after a clean import; target text remains readable at the specified display size; loop boundary is inaudible under the intended playback conditions; required localized strings are present; units match the destination contract. Use project-specific thresholds only after a target has been chosen. There is no universal safe polygon count, texture size or loudness target for all media.

Aesthetic checks still matter, but label them as review judgments. "Consistent with the approved reference sheet" is more useful than "high quality." Record the reviewer, reference version and scope rather than converting taste into an unsupported numeric certainty.

Identify disqualifiers and unknowns

A family likeness to an unlicensed character, uncertain voice consent, unreadable text, an irrecoverable source dependency or absent rights evidence can block use regardless of visual appeal. Unknown is a valid state. Distinguish "not checked," "not applicable," "checked and failed" and "checked and passed"; blank cells should not silently mean success.

Original counterexample: two renders look identical, but one source retains real text and the other paints the text into a texture. Both may pass an appearance review. Only the first necessarily supports accurate rapid localization. The contract decides whether that difference matters.

A pipeline with independent evidence gates

The following lifecycle is a proposed production model. It describes records and decisions, not permission to automate Blackwish or deploy anything.

Brief -> acquire or create -> normalize -> edit -> derive -> validate -> review -> authorize -> publish -> retain or retire. A failure can return work to an earlier stage. Every transformation creates a new version or derivative relationship rather than silently replacing its source.

Separate four kinds of readiness

Creative readiness: the asset supports the intended design, tone, meaning and audience. **Technical readiness:** a named version passes named checks in a named target. **Rights readiness:** the evidence supports the proposed use, distribution and modifications. **Authority readiness:** the responsible person has explicitly authorized the specific action. These dimensions should remain separate even when a dashboard presents a combined release decision.

An asset can be technically valid but rights-blocked, rights-cleared but inaccessible, or creatively approved but incompatible with a target engine. A generated draft may remain valuable for internal comparison while being unsuitable for external distribution. Do not rename all of these cases "approved."

Freeze the input and record the transformation

Keep a stable source identifier and content digest, tool and plug-in versions, settings, dependency versions and output digest. Preserve original scans, recordings, drawings and purchased packages separately from normalized working copies. Where storage is constrained, define a deliberate retention policy instead of deleting originals because a preview looks good.

A digest proves byte identity under its algorithm, not authorship, authenticity, truth or lawful use. A transformation log can be incomplete or false. Pair machine records with the actual evidence needed for the claim: source files, license text, invoices or consent records where applicable, and target-specific test results.

Validate the package rather than only the central file

Open the export in the receiving application or a representative clean environment. Check textures, fonts, linked media, shader behavior, animation, coordinates, units, names and relative paths. A file that opens on the creator's computer may merely be resolving dependencies from an unrecorded cache. Review a release candidate from its packaged location rather than the original authoring directory.

Make stale evidence visible

Editing a mesh invalidates some geometry tests; changing a translation invalidates some timing and layout reviews; changing a license or target distribution invalidates some rights conclusions. Mark affected evidence stale. Reuse unaffected evidence only when its dependency relationship is clear. An old "pass" should never travel indefinitely with a new asset merely because its name stayed the same.

Stopping condition: if a required gate is unknown, failed or stale, keep the candidate in a reviewable state and name the smallest missing evidence. Do not substitute more generations for the missing check. Release is a decision about a particular package and use, not an intrinsic property of a file.

Concept art and two dimensional images

Concept art should answer a design question, not merely produce an attractive picture. A style sheet establishes choices such as silhouette, proportion, palette, material language and detail density. A production model sheet must additionally resolve how views agree. A mood board can remain suggestive; an authoritative turnaround cannot leave the back of an object to accidental invention.

From exploration to a reliable reference

Start with intended use and a bounded design vocabulary. For an ordinary delivery bicycle, specify whether the cargo box is a fixed front container or a rear rack; record wheel size and overall proportions if they matter. Explore several silhouettes, select one direction, then construct a rough common model or aligned drawing system before producing front, side and rear references. Label unresolved features rather than hiding ambiguity in painterly detail.

Hand drawing offers direct control. Photography supplies observed detail but carries capture, privacy and rights considerations. Procedural layouts are useful for repeated formats. Generative images can widen exploration, but cross-view consistency, exact text, small mechanisms and requested edits still need independent review. Upscaling can improve presentation without recovering factual detail that was never captured.

Keep the source richer than the export

Preserve layers, masks, selections, editable typography, vector components and color settings when revision matters. Derive crops and sizes from a common source. A flattened delivery image can be appropriate, but should not silently replace the master. PNG provides lossless raster storage with optional alpha and color information; this does not mean every application interprets every color-related feature identically. [V08]

Review at actual display or print size as well as magnified. Check edge quality, readable labels, intentional crop, visual hierarchy and sufficient context. A small educational diagram can fail because its arrows are ambiguous even when its rendering is beautiful. Check light and dark backgrounds for fringes around transparency; alpha assumptions are part of the contract, not a cosmetic afterthought.

Original failure fixture

A generated picture of a museum trolley shows four wheels in one view and three in another. Both images are plausible individually. The failure is relational: they do not specify the same object. The appropriate repair is a coherent blockout and corrected views, not an average similarity score or a larger batch of unrelated candidates.

Acceptance evidence: approved reference revision; intended dimensions or explicit stylization; final-size preview; color and transparency checks; factual-text review; rights evidence; and a preserved editable source. For concept-only material, mark the limitations clearly so it cannot be mistaken for a measured production specification.

UI icons sprites tiles and typography

These assets are small but system-dependent. An icon has states and meaning; a sprite has timing and a pivot; a tile belongs to a larger pattern; typography must shape real language. Evaluate them as families rather than isolated decorative images.

Icons and interface graphics

Start from semantic roles: information, navigation, confirmation, warning and unavailable state. Keep a consistent visual grammar and provide non-color distinctions where meaning depends on state. Review at the smallest supported size, in light and dark themes, and with focus, selection and disabled treatments. WCAG's non-text contrast guidance specifies a 3:1 contrast requirement for applicable visual information against adjacent colors, with exceptions; it is not a universal rule for every decorative mark. [V19]

SVG can contain vector and raster material, styling and dynamic behavior. It is not simply a safe bag of paths. Test the actual destination's supported subset, including clipping, gradients, strokes, text and embedded resources; treat imported SVG as untrusted content until appropriately handled. Preserve editable source even if the destination needs rasterized variants. [V07]

Sprites and tiles

Define logical canvas, pivot, frame order and duration before drawing a sequence. When frames are trimmed for packing, retain offsets; otherwise an object can jitter despite individually correct pictures. Atlas padding and edge extrusion should be tested with actual filtering and mip levels. A seemingly harmless neighboring sprite can bleed into another at reduced size.

For tiles, review edges, corners, transitions and a large representative area. Seamless edges do not prevent conspicuous repetition. Use controlled variation that preserves the design language and connection rules. A pathway tile set should distinguish visual joins from traversability; a matching picture of a path does not establish a valid route in an interactive scene.

Typography and language

A font is a software asset as well as a visual choice. OpenType supports substitution, positioning and variation; one character does not always correspond to one independently positioned glyph. Test real strings, combining marks, ligatures, numerals, punctuation, fallback and required scripts. Preserve shaping relationships when making a subset. [V20]

Outlining display lettering can stabilize appearance but removes live-text editability and search semantics. Keep a live-text master and provide semantic text where required. Verify the actual font's embedding and modification terms rather than assuming that download availability grants all uses. Localization needs layout flexibility, not just a larger English text box.

Original counterexample: a transit icon works in a large mockup but becomes indistinguishable from "pause" at actual size. Its vector construction and perfect export do not rescue its meaning. The acceptance check must include the user's real viewing context.

Three dimensional geometry and props

Choose a representation for the task. Parametric or CAD authoring supports controlled dimensions and relationships; polygon modeling offers direct surface control; sculpting supports complex organic detail; scanning and learned generation can supply starting points. None removes the need to decide which parts must move, deform, collide or remain editable.

A practical geometry workflow

Block out proportion and silhouette before detail. **Refine** meaningful forms and separate moving parts. **Retopologize** when the source topology is unsuitable for deformation, shading or runtime cost. **Unwrap** with an explicit purpose for each UV set. **Bake** details into maps when that preserves the required appearance at lower cost. **Export and inspect** in the receiving tool.

Topology is not a universal contest between triangles and quadrilaterals. A rigid book and a bending sleeve have different needs. Inspect degenerates, unintended internal faces, holes, self-intersections and surface orientation according to the target contract. An open boundary can be intentional in a render-only asset; the same boundary can be a serious problem in a physical-output workflow.

Units and transforms are content

Declare units, up and forward axes, handedness, origin, pivot and expected bounds. glTF specifies a right-handed system, positive Y up and meters for linear dimensions. OpenUSD's stage-wide linear-unit metadata has a centimeter fallback when unauthored, and assembly may require correction when assets use different units. Setting a label is not the same as transforming all geometry, joints, animation and cameras correctly. [V02, V03]

Use a measured reference object and verify final bounds after parent transforms. A tiny mesh under a large parent scale can look correct while producing surprising collisions, deformation or export. The pivot should serve the intended operation: a hinge rotates around its hinge axis, not whatever origin happened to survive modeling.

UVs and baking

Balance distortion, texel density, seams, reuse and material assignments. Overlap may be intentional for a repeating material but unacceptable for a particular unique bake. Record which UV set serves which purpose. Inspect checker patterns and final textures at seams, undercuts and thin surfaces. Bake projection can accidentally capture neighboring geometry; later triangulation or normal changes can invalidate a previously good result.

Original failure fixture: a generated coffee grinder depicts its adjustment knob as painted surface detail. That may be adequate for a distant illustration, but it is not an independent rotating component. The contract determines whether to accept the appearance, reconstruct the knob or reject the candidate.

Keep: the dimensioned or high-detail master, final topology, bake recipe, maps, export profile and target evidence. A low-resolution runtime mesh should not become the only source for future editions.

Materials textures lighting and color

A texture is sampled data. A material describes how a surface responds to light. A lighting setup determines how that response is seen. Confusing these layers produces attractive previews that fail when reused.

Preserve channel meaning

Distinguish base color, roughness, metallic, normals, opacity, emission, displacement and occlusion. Treat numerical maps as data rather than photographs; applying an unintended color transform can change their meaning. Record channel packing and normal-map conventions explicitly. A map with the correct filename can still contain reversed channels or a different tangent convention.

A photograph of wood includes illumination. Reusing it unchanged as base color can bake shadows into an object that will be lit again. A proposed authoring route is reference or capture, material interpretation, editable graph or layered maps, then neutral-light comparison. PBR describes a modeling approach, not a guarantee that any chosen maps are realistic or mutually consistent.

Compare across conditions

Use fixed cameras and a neutral review scene, then check materially different lighting conditions. Inspect highlights at grazing angles, roughness variation, surface scale, seams and displacement bounds. A wood grain that is plausible on one object can look enormous on the adjacent one. Repeating maps should be checked over a large area, not only at their boundary.

MaterialX specifies material-graph elements and node families, helping exchange descriptions. It does not promise that every renderer supports every node or produces identical images. Where a graph must be baked, document what is lost: view dependence, procedural editability, displacement detail or renderer-specific effects. [V06]

Make color management a dependency

Record source interpretation, working space, output transform and relevant configuration version. OpenColorIO provides a production color-management framework across applications; its presence alone does not establish that a particular scene is configured correctly or that a display is calibrated. Keep data channels distinct from display-referred artwork. [C04]

For compositing, record alpha convention and image windows. OpenEXR's technical introduction explains premultiplied-color conventions and the distinction between data and display windows. A cropped render can be correct yet positioned incorrectly if those windows are ignored. Inspect edges against contrasting backgrounds and avoid repeated lossy display conversions of working material. [V10]

Lighting is an authored asset too

Keep lights, exposure, camera, render settings, useful passes and compositing decisions when future revisions matter. Review sequences for flicker and temporal noise, not only a hero frame. A dramatic final light can conceal surface defects; a neutral inspection light can expose them without being the desired final aesthetic.

Counterexample: a concrete texture passes a seamless-edge test but repeats the same painted shadow every few meters. The tiling algorithm succeeded; the reusable-material contract failed. Repair the source or narrow its permitted use.

Characters rigs and deformation

A character is a coordinated set of appearance, structure, motion, voice, text and identity decisions. An attractive neutral pose is only one narrow test. Start with what the character must do and which traits must remain stable across media and revisions.

Separate the control rig from exported data

The animator's control rig can include constraints, inverse kinematics, drivers and custom logic. A destination may accept only a skeleton, skin weights and baked animation. Exporting a skeleton does not preserve the whole authoring interface. Establish neutral pose, hierarchy, joint orientation, scale, root behavior and attachment conventions before producing many clips.

OpenUSD's UsdSkel schemas distinguish skeletons, bindings, joint animation and blendshapes, including mapping between joint orders. This supports a useful conceptual separation, not a promise that any two character systems exchange every control or deformation feature. [V18]

Design for the required movement

For a park worker who points and carries a watering can, test shoulder rotation, elbow bend, wrist orientation, hand contact and clothing clearance. Skin weights should follow intended influence and be checked through the real motion range. Corrective shapes can repair pose-dependent loss of volume or intersections, but they add dependencies and must survive the target pipeline.

Blendshapes require stable correspondence. Changing vertex order after authoring can break expressions even if the neutral mesh looks unchanged. Test required combinations, not only one shape at a time. A smile plus a blink can reveal an intersection that neither produces alone. Record the mapping and range expected by lip-sync or facial-animation systems.

Variation needs protected interfaces

A character family can vary clothing, proportions or face shape while keeping approved identity and technical interfaces stable. However, a longer arm can invalidate an old grip, and a wider sleeve can collide during an unchanged gesture. Randomization should create new candidates requiring relevant checks, not presumed equivalent members of a certified family.

Capture and reference material also have rights boundaries. Permission to record a motion is not automatically permission to reuse a person's likeness, clone their voice or generate unrelated performances. Keep technical motion data and consent evidence linked but separately protected. Do not infer identity clearance from a rig's successful import.

Original counterexample

Two variants share the same skeleton names and pass structural validation. One has different shoulder proportions and collapses when reaching overhead. Name compatibility is evidence of a mapping, not proof of correct deformation. The repair may be weights, proportions, corrective shapes or a different animation; adding more idle clips does not address the failure.

Acceptance evidence: reference-sheet consistency, required pose suite, contact checks, facial combinations, attachment behavior, target import and playback, visibility bounds, and explicit unresolved limitations. Keep the source rig even when delivery requires baking its behavior.

Animation motion simulation and timing

Animation can be hand-keyed, captured, procedural, simulated or generated as images. These are not interchangeable outputs. A video of someone opening a door is not necessarily a retargetable motion clip, an editable rig or a sequence with usable event markers.

Keyframes and motion capture

Hand-keyed motion offers deliberate control over poses and timing. Capture provides observed performance but requires calibration, solving, cleanup, retargeting and contact repair. A proposed capture-to-clip workflow retains authorized raw material, solves motion, removes outliers, adapts proportions, fixes foot sliding and hand contact, then checks the actual target character.

Define root motion, in-place behavior, clip duration, timebase, interpolation and loop intent. Inspect start and end frames, velocity continuity and transitions to other clips. A cycle can have matching endpoint poses yet visibly pop because velocity changes abruptly. A walking clip can look correct in isolation while moving at the wrong speed relative to world translation.

Events belong to meaning

A footstep should align with contact and the surface, not simply a copied frame number. Retargeting, speed changes, altered footwear or a new gait can invalidate old sound timing. Keep event IDs and attachment names versioned with the source. The same principle applies to object pickups, pointing cues, captions, camera cuts and interaction windows.

OpenTimelineIO represents editorial timelines through tracks, clips, transitions and references to media. Its documentation notes that effects are application-specific and that a referenced range can exceed available media without automatic verification. Therefore an editorial interchange file is not proof that all media exist or every effect survives. Package and inspect the referenced assets separately. [C03]

Simulation and baked representations

A simulation needs scale, initial conditions, solver settings, time step, constraints, cache range and tool version. A low-resolution preview is not proof of final behavior. For an ordinary cloth banner, inspect attachments, collisions, bounds and temporal stability. Upstream changes should mark caches stale rather than silently reusing them.

Choose runtime simulation only when its cost and variability fit the experience. A baked cache or flipbook can be more predictable for a fixed sequence, but gives up interaction or viewpoint freedom. Preserve the original recipe where future changes matter. A physically inspired model can be useful artistically without being a scientific prediction.

Original failure fixture

A museum guide's pointing animation is retimed, but the spoken word "here" and highlight cue stay at their old timestamps. Every individual file is valid; the combined sequence is wrong. The acceptance test must run the complete timed scene and include pause, interruption, skipping and resume behavior if the destination supports them.

Stop when output type does not match the needed controls. Do not promise editable motion from flattened generated frames without a separate reconstruction and validation plan.

Modular environments and world assembly

A modular kit is a set of compatible interfaces, not merely many related-looking objects. Establish grid, pivot, dimensions, connection rules, material scale and collision policy before expanding the library. An attractive wall segment that cannot join its neighbors is not a successful module.

Build a grammar before generating volume

For a community reading room, define walls, floor bays, shelves, doors, windows and furniture zones. Separate required landmarks and circulation regions from optional decoration. Each module should state connection types and orientation, not rely on an artist guessing from its name. Keep the conceptual layout distinct from the meshes that depict it.

Grammars express repeated composition. Constraints express what must remain true across the whole result. A shelf can fit its local bay while blocking the room's only route. Local geometric compatibility does not establish global access, readable navigation or plausible use. Validation should check both the modules and their assembled context.

Reuse with deliberate variation

Instancing can share data across repeated objects. OpenUSD's documentation explains shared scenegraph prototypes and restrictions on editing within instances; generated prototype paths are not stable asset identifiers. Retain stable conceptual IDs and record intentional per-instance differences outside incidental ordering. [V05]

Variation should be hierarchical: building family, module, surface condition and local detail. Randomizing everything independently often erases coherent design. For a park, protect paths and recognizable landmarks, vary planting within declared regions, and record density and exclusions. More nominal combinations do not necessarily create more meaningful diversity.

Terrain and vegetation

Separate macro layout, elevation, materials, collision and navigation. Noise can create irregularity, but does not guarantee plausible drainage or usable slopes. Preserve scale before adding detail. For vegetation, coordinate species silhouette, branching, leaf treatment, placement and wind behavior rather than treating every property as unrelated randomness.

A visually convincing tree is not a biological identification. A building façade grammar is not structural engineering. These distinctions matter when the same asset moves from fictional scenery into educational, architectural or public-information material. Carry the intended use and uncertainty into the derivative.

Original failure fixture

A generated plaza obeys every paving adjacency rule and places benches without geometric intersections. It nevertheless blocks the route from an entrance to the exhibit. The correct diagnosis is a missing global constraint, not insufficient texture variety. Stop and revise the contract rather than approving the scene because its local pieces look polished.

Acceptance evidence: kit interface sheet, representative joins and corners, assembled-scene review, intended circulation checks, collision and culling review, variation stability under small edits, and target performance. A kit should be judged by representative assemblies, not only a contact sheet of isolated parts.

Scanning photogrammetry and view based representations

Capture is valuable when an existing surface contains detail that would be expensive to recreate. It is not automatic truth or automatic permission. Decide what is being captured, why, who authorizes it and what claims the resulting asset may support.

Photogrammetry as a pipeline

COLMAP distinguishes structure from motion from multi-view stereo and provides guidance on image conditions. A practical capture plan needs informative texture, overlapping coverage, suitable exposure and stable subjects. Shiny, transparent, moving, featureless and occluded regions deserve particular skepticism. Record scale references and calibration information when dimensional claims matter. [V16]

A proposed workflow is authorized capture, preservation of originals, reconstruction, inspection of holes and outliers, scale establishment, cleanup, retopology where needed, UV and material work, then comparison with independent dimensions or unused views. A sharp texture can hide poor geometry. A reconstruction that looks right from its source cameras can fail from a new angle.

Meshes are not the only three dimensional representation

A point cloud samples spatial points. A mesh provides explicit surfaces and topology. A radiance field models scene appearance for view synthesis. NeRF's original work demonstrates a learned radiance-field approach to novel views; it does not make the resulting representation equivalent to a hand-edited mesh with rig, collision and manufacturing semantics. [V13]

Gaussian splatting uses optimized three-dimensional Gaussians to represent captured appearance. The original research reports results for its tested conditions; those results are not a blanket frame-rate guarantee for arbitrary scenes or devices. A beautiful walkthrough representation can still lack editable surfaces or reliable unseen regions. [V14]

Choose what the destination actually needs

For a guided visual tour, captured appearance may be the primary value. For a movable prop, relighting, part separation and collision may dominate. For a museum interpretation, distinguish measured evidence from inferred reconstruction. For physical printing, a separate manufacturable representation and process review are necessary.

Conversion may involve extracting geometry, remodeling, simplifying and baking. It can lose detail or introduce new assumptions. Record that transformation rather than claiming the final mesh was directly observed in every region. Keep raw captures and an uncertainty note when future interpretation matters.

Original failure fixture

A captured display cabinet looks excellent from the front. Behind it, the reconstruction invents or omits surfaces that were never photographed. The asset may be acceptable for a fixed camera but not for unrestricted inspection. The honest repair is either additional authorized capture, explicit reconstruction or a narrower supported viewing envelope.

Acceptance evidence: capture provenance, permission scope, coverage map, scale method, known weak surfaces, chosen representation, cleanup history and destination-specific review. Absence of obvious artifacts in one view is not evidence that all views or measurements are reliable.

Procedural systems rules graphs and constraints

Procedural generation earns its complexity when many assets share stable relationships. A parameterized bench can vary length while preserving joint structure; a layout grammar can assemble rooms; a placement system can distribute approved objects. Define controllable design variables separately from incidental randomness.

Choose the correct layer

A modeling graph transforms geometry. A build graph manages dependencies and work. A scene-population graph places or modifies content in context. SideFX's PDG documentation describes work items, dependencies and selective recomputation; Unreal's PCG overview describes spatial data and graph-driven content generation. These are examples of different layers, not interchangeable tools or recommendations to adopt a particular vendor. [P02, P03]

Typed, bounded inputs should reject impossible combinations clearly. A shelf generator asked to fit more compartments than its declared dimensions allow should report the conflict rather than quietly reducing shelf thickness. Keep examples of valid, invalid and boundary inputs alongside the recipe. A successful evaluation only means the graph ran; it does not prove the requested constraints were satisfied.

Local compatibility is not global validity

Wave Function Collapse provides example-driven pattern or tile generation. A road map can satisfy every local adjacency while leaving disconnected islands. Constraint-solving tools such as OR-Tools can support explicit feasibility problems, but a project must still define variables, domains, constraints, timeouts and what a solver result means. Feasible is not necessarily optimal; a timeout is not proof that no solution exists. [P05, P04]

For an ordinary museum floor plan, separate module matching, reachability, protected spaces and visual composition. Do not infer building-code or accessibility compliance from a visual layout algorithm. A fictional scene and a real installation have different evidence requirements.

Reproducibility needs more than a seed

Record recipe, dependency and tool versions, normalized inputs, ordering rules and relevant environment. PyTorch explicitly cautions that identical seeds do not guarantee reproducibility across releases, platforms or CPU/GPU execution. The broader design lesson is to preserve actual outputs and define the promised reproducibility level. [P19]

Use **byte-identical** only when demonstrated in a specified environment; **tolerance-equivalent** only with stated comparison bounds; and **traceable but not reproducible** when the preserved result and its record are the reliable evidence. Stable subcomponent IDs help prevent a small edit from reshuffling an entire scene. Do not use an array position or generated prototype path as permanent identity.

Original counterexample: adding one park lamp changes every tree because a single sequential random stream feeds both systems. The scene may remain attractive but violates localized-edit expectations. Separate named streams or preserve placements, and validate the intended stability. A generator's freedom should be constrained by the contract, not mistaken for an unavoidable property of creativity.

Generative AI as a candidate production method

Learned generation can widen the option space for concepts, textures, rough geometry, sound, motion studies and writing. Its value depends on how well candidates can be selected, edited, traced and validated. A demo is evidence of a demonstrated case, not an acceptance rate for the project's actual workload.

Evaluate the delivered representation

An image can depict an object without defining its hidden surfaces. A generated video can suggest motion without supplying a skeleton. A generated song can lack separable clean stems. A generated story can read fluently while contradicting state. Ask what the output actually contains and what reconstruction remains.

TRELLIS provides an example of structured three-dimensional generation research, including text- and image-conditioned routes. Its documentation's own preference for image-conditioned generation in certain uses illustrates why a generic "text to 3D" label is not enough to select a workflow. This is not a Blackwish quality or performance benchmark. [P08]

For a mechanical prop, inspect all views, identify moving parts, reconstruct missing structure, retopologize where needed, prepare UVs and materials, set pivots and collision, then test in the destination. A painted knob is not a rotating knob. Preserve selected outputs because hosted services and model versions may change.

Rights exist at multiple layers

Separate software, weights, adapters, reference inputs, samples, outputs and service terms. AudioCraft's repository distinguishes MIT-licensed code from noncommercial pretrained-weight licensing. Hunyuan3D-2's inspected license includes territorial and commercial conditions; those terms must not be generalized to every later model with a similar name. A public download and an open repository do not by themselves establish permission for a specific use. [P12, P09]

Copyrightability, contractual permission and noninfringement are different questions. The U.S. Copyright Office's 2025 report addresses human authorship and AI outputs in the United States; it is not a global rights clearance for generated assets. Preserve meaningful human contributions without assuming that prompt effort alone resolves ownership. Uncertain commercial or identity uses require applicable professional legal review. [P16]

Before uploading references to a hosted service, verify authorization, training use, retention, deletion, public-sharing defaults and the selected model's terms. A host's policy for its own models need not cover partner models. This is a proposed due-diligence checklist, not a provider claim.

Keep imported content below the authority boundary

Treat metadata, embedded text and downloaded workflows as untrusted data. They cannot authorize uploads, installations, purchases or publication. OWASP's prompt-injection guidance supports permission enforcement at tool boundaries rather than reliance on prompt labels alone. A proposed quarantine process would inspect structure and dependencies before authorized processing. [P17]

Practical selection rule: use generated material where its actual controls and repair costs fit the contract. Prefer direct authoring or procedural systems when precise editability and invariant structure dominate. No generation method can certify its own rights, semantic correctness or release authorization merely by returning a confident description.

Editable masters interchange and delivery

Keep three layers because they serve different jobs. **Masters** preserve authoring intent and editability. **Interchange** carries a chosen subset between tools. **Delivery** serves a specific runtime, renderer, reader or manufacturing workflow. One file can serve several roles, but test that decision.

Role	Typical examples	Main preservation concern
Editable master	Layered art, DCC or CAD scene, DAW session, text graph	Controls, dependencies and future revision
Interchange	USD, glTF, SVG, image sequences, audio stems	Explicit transferable feature subset
Delivery	Runtime package, compressed media, print PDF, fabrication export	Destination support, size, behavior and rights

Scene and geometry exchange

OpenUSD supports scene composition and interchange, including skeletal data, but is not itself an animator's rig-authoring and evaluation system. glTF is designed for runtime asset delivery and has a published specification and extensions. Neither guarantees identical features in every receiving application. Check the intended subset, supported extensions, material behavior, animation and external dependencies. [P20, V01]

Specify spatial conventions, units, names, pivots, timebase and package paths. Avoid treating a successful import as proof of visual or behavioral fidelity. A missing material extension can yield a valid file that looks wrong in the target.

Appearance and media exchange

Materials may need graphs, baked maps or both. Images need declared color and alpha interpretation. Audio needs channel layout, sample-rate policy, loop and timing information, with separate stems when later mixing matters. Text needs semantic structure, stable IDs and localization context. An interchange format's ability to store a feature does not prove that both exporting and importing implementations preserve it.

Keep a manifest of fonts, textures, linked scenes, recordings, caches and plug-ins. A package that only works through the author's local cache is not portable.

Round trips are targeted tests

Choose the properties that matter, export, import into the actual receiving application, then compare those properties. Do not demand a byte-identical round trip when the format intentionally changes representation; do not accept a visual match when editability or metadata is the requirement. Record known losses and acceptable fallbacks.

Original counterexample: a bench exports correctly, but the receiving scene assumes a different unit and applies a hidden parent scale. Its preview looks normal after manual adjustment while collision remains wrong. The handoff should verify dimensions and behavior, not only the screenshot.

Retention rule: keep the richest practical source and the exact delivered package. Rebuilding an old release by rerunning a changed generator is not a reliable rollback strategy. A conversion creates a derivative; it does not retroactively improve the provenance or permissions of its inputs.

Music effects ambience and audio delivery

Sound assets combine content with behavior. A musical cue needs composition, recordings, stems, timing and transitions. An effect needs a meaningful event and perspective. An ambience needs controllable layers and a sensible repetition policy. A pleasing file is not yet an integrated audio system.

Choose an authoring route by control needs

Composition and performance offer deliberate identity and timing. Recorded libraries supply reusable detail but require license and perspective matching. Procedural synthesis suits repeatable interface cues, machinery and controlled variation. Generative audio can suggest textures or drafts, but exact timing, unwanted speech, clean separation and revision stability need review. Preserve raw recordings, sessions, patches and edits rather than only the final mix.

Music contains separate composition and sound-recording layers; the U.S. Copyright Office describes them as distinct works. Inventory both rather than assuming one permission settles all reuse. This is a starting point for rights review, not jurisdiction-independent legal advice. [A01]

Design adaptive music deliberately

Vertical layering changes compatible parts; horizontal resequencing moves between sections. Align stems to a common origin, preserve intended silence and decide where reverb tails live. Stem separation from a finished mix may contain leakage and is not equivalent to original multitracks. Wwise documents source and destination cues, transitions, fades and post-exit behavior; these mechanisms still require composition and listening review. [A02]

An original museum sequence might move between arrival, browsing and closing announcements. Test rapid changes, pauses, interruption, repeated events, long idle periods and speech-heavy moments. A set of excellent tracks can clash when crossfaded. Critical information should not be delayed merely to finish a musical phrase.

Build a useful sound taxonomy

Classify effects by action, material, force, distance and environment. A few meaningfully different variants can be more useful than many near-duplicates. Keep diffuse ambience separate from localized events. Review loop seams and longer-term repetition: a seamless loop can still repeat a conspicuous bird call. Exclude unintended intelligible conversations or background music unless appropriately cleared.

Derive for the actual destination

Godot's documentation illustrates format-dependent storage, decoding and loop considerations for WAV, Ogg Vorbis and MP3. Other engines and versions have their own constraints. Do not choose a codec by extension alone or impose one sample rate and loudness number on every use. [A05]

EBU R 128's -23 LUFS recommendation is programme-level broadcast guidance, not a universal normalization target for every game effect or music stem. Check the release destination and measure the appropriate complete output. Review clipping, true peak, intelligibility, channel routing, fold-down, tails and aggregate mix behavior. [A06]

Stop condition: missing rights, unintended speech, broken event timing or an unacceptable mix remains a blocker even when the file decodes successfully.

Voice localization and accessible equivalents

A dialogue line is a linked family: approved text, locale, pronunciation, performance, timing, captions, optional facial cues and rights evidence. Changing one member can invalidate the others. Stable IDs should survive changes in wording, while versions identify the actual content.

Voice identity and performance

Describe performance through pacing, energy, articulation and emotional intent rather than using a famous person's identity as a shortcut. For synthetic voice, use a properly licensed voice or an explicitly authorized identity within the applicable platform's rules. ElevenLabs' inspected Professional Voice Cloning documentation requires own-voice verification; permission from someone else does not authorize bypassing a platform's specific workflow. [A04]

As a proposed production control, distinguish permission to record, edit, synthesize, localize, generate new dialogue and distribute in named media and contexts. Keep sensitive consent evidence protected rather than embedding it in public files. Identity verification, performer approval and rights clearance are different checks.

Review final lines against the approved script, especially names, negation, numbers and conditions. Maintain pronunciation notes. A voice that sounds fluent can say the wrong word. Mark temporary narration clearly so it cannot quietly become a release dependency. A final voice change can affect animation, captions, camera pacing and music simultaneously.

Localization is structured adaptation

Supply translator context: speaker, recipient, situation, visible object, tone, placeholders and constraints. Avoid constructing sentences by concatenating translated fragments. ICU's message-format guidance supports language-dependent plural selection and structured formatting; use the exact syntax supported by the chosen runtime and translation tools. [A11]

Test placeholder preservation, markup, glyph coverage, fallback, right-to-left layout, expansion and actual UI sizes. Localized speech need not match the original duration. When timing is constrained, plan adaptation and retiming instead of forcing unnatural delivery. Font licenses and coverage remain part of the asset package.

Captions transcripts and description do different jobs

W3C's guidance distinguishes captions for speech and relevant non-speech audio from descriptions of important visual information. Plan both with the media. A transcript does not substitute for synchronized captions when timing matters, and captions do not explain an essential silent visual action. [A13, A14]

Microsoft's XAG 104 supplies practical subtitle and caption guidance including speaker or directional information and configurable presentation. Adding text alone does not certify comprehensive accessibility. Review the actual experience, including introductory content, controls and important information when sound is unavailable. [A15]

Original counterexample: a caption identifies a speaker whose identity is intentionally unknown at that point. Its transcription is accurate but its information boundary is wrong. Use a context-appropriate label and retain the intended knowledge state. Accessibility assets must preserve meaning without silently rewriting the story.

Narrative text and educational assets

Narrative assets include dialogue, descriptions, tutorials, branching scenes, journals, reference entries and interface messages. Their shared problem is consistency across context and revision. Fluent prose is not evidence that the underlying facts or transitions are valid.

Separate the layers of truth

Maintain distinctions among approved world facts, a character's beliefs, audience knowledge, interaction state and displayed wording. A character can intentionally be mistaken without corrupting the underlying record. A generated line should not silently change rewards, reveal a secret, invent a factual claim or authorize an action because it sounds plausible.

For an original neighborhood-history exhibit, a volunteer may know when a building opened but not who designed it. A draft that names an architect is a new claim requiring evidence, not a harmless stylistic flourish. In interactive fiction, a true fact can still be unavailable to a character at that stage. Record both truth and knowledge eligibility.

Structure before surface prose

Ink is an interactive-narrative language with distinct authoring, compiler and runtime components. Yarn Spinner's localization and asset guidance links line IDs with text and media. These demonstrate useful structures; neither automatically supplies coherent stories or accurate translations. Keep canonical facts, conditions and allowed transitions separate from verbal variants. [A08, A10]

Use stable IDs, source revisions, context, prerequisites and dependency references. Validate unreachable nodes, missing exits, contradictory conditions, absent fallbacks, repeated rewards and reload behavior where applicable. A deliberate ending differs from an accidental dead end. Test boundary states, not only a preferred walkthrough. No such software test or human experiment was performed for this reference.

Design educational assets for evidence and access

Preserve real text, headings, reading order, semantic links and the underlying structured data of diagrams. W3C's complex-image guidance supports a concise identification plus a fuller representation of essential information. A generic "diagram" label cannot convey the relationships a learner needs. [A17]

Keep claim-level sources, intended learning objective, prerequisites, answer rationale and uncertainty. Distinguish an invented example from a real-world assertion. A stylized map can be suitable for teaching a concept while being unsuitable for navigation. A historical reconstruction should label inferred details rather than present plausible generated completion as documentary evidence.

Original failure fixture

The prose says a museum ticket permits re-entry, while the interaction state denies it. Both text and software can be internally well-formed; the cross-asset contract is inconsistent. The remedy is to consult the authoritative policy and update the affected versions, not choose whichever artifact is more polished.

Acceptance evidence: source and fact review, state and reference checks, reading-context review, localization dependencies, accessible alternatives and explicit unresolved interpretations. Generated text stays a proposal until the appropriate reviewer accepts the actual version and its intended use.

Games interactive web VR and AR

Interactive assets must work as part of a changing scene under bounded resources. Define a target profile before optimization: device class, display conditions, memory, loading, rendering features, interaction and accessibility. There is no universal polygon count that makes every asset efficient.

Optimize the measured bottleneck

Consider geometry, vertex splits, material slots, draw calls, shader complexity, transparency, shadows, animation, collision, texture residency, decoding and streaming. A small compressed download can expand into large runtime memory. A low-polygon object with expensive transparency can cost more than a denser opaque one. Profile a representative scene rather than inferring performance from a file statistic.

Level of detail should reduce cost as visible importance changes while preserving silhouette, motion and material continuity. Inspect transition popping and bounds. Atlases can reduce material changes but waste space, complicate streaming and introduce filtering bleed. Instancing can reduce duplication while limiting independent edits. These are tradeoffs, not automatic improvements.

KTX supports GPU texture data and mipmaps; KTX 2 includes supercompression and transcoding mechanisms. The actual runtime must support the chosen combination. A container suffix does not establish device compatibility, memory use or acceptable visual loss. [V12]

Structural validation is necessary but narrower

Khronos's glTF Validator checks format-level issues. A passing result does not establish visual fidelity, legal permission, useful collision, correct deformation or acceptable performance. Follow with target import and scenario-based review of the actual delivery package. [V11]

Define missing-asset behavior and supported fallbacks. A high-end material extension might need an acceptable simplified version on another target. Do not silently substitute a fallback that changes important meaning, such as removing a state indicator or making a transparent barrier appear solid.

VR and AR add spatial obligations

Check actual scale, stereoscopic views, near-field detail, interaction bounds and sustained frame timing on the intended device. Flat-screen screenshots do not establish stereo correctness or user comfort. A fixed frame-rate claim must identify hardware, scene, settings and measurement conditions. This reference does not perform user testing or certify comfort.

In AR, consider alignment and occlusion expectations, lighting mismatch and how essential content remains readable against changing backgrounds. Physical-space interactions require separate contextual and safety review; a plausible digital placement is not authorization to alter a real environment.

Original failure fixture: a park scene runs well from its opening camera but stalls when all transparent leaves fill the view. The average from the easy angle is misleading. Define representative worst-relevant views and actions, record frame-time distributions and memory peaks, then retain the exact package and settings that were assessed.

Film books and physical output handoffs

Different destinations justify different derivatives. Preserve the shared identity and source relationships, but do not force one optimized representation to serve every medium.

Film and offline imagery

Retain useful scene composition, high-detail geometry, displacement, lights, cameras, render passes and reproducible color settings. Longer rendering time can be acceptable, but it is not unlimited. Estimate a sequence using representative complexity rather than multiplying one simple frame. Review motion blur, noise, temporal flicker, edge contamination and compositing interactions across time.

For a product or museum film, the close-up model may require detail irrelevant to a game background. Keep that high-quality master separate from the runtime derivative. A flattened rendered sequence is a valid delivery artifact but not a replacement for an editable scene when future camera or lighting changes are expected.

Books and print graphics

Work backward from physical size, printer requirements, bleed, safe areas, color/output profile and font handling. As a simple arithmetic example, an eight-inch-wide image at 300 pixels per inch needs 2,400 pixels across. Changing a resolution tag alone does not create those pixels. The printer's actual requirements take precedence over that illustrative number.

Keep text semantic and editable where possible. Preserve source layout and linked images rather than only a flattened page image. Review the final physical size, reading order, caption relationships and print proof conditions. Reflowable books and fixed-layout books have different constraints; a layout suitable for a large page may be unreadable on a small screen.

Physical three dimensional printing

Convincing model is not manufacturable print, and manufacturable print is not a qualified functional object. Preserve editable source, units, revision, licenses, original generated output and repair history. Watertightness is only one geometric property; it does not establish wall thickness, fit, support access, trapped-material removal, durability or safety.

3MF is a manufacturing-oriented format with a core specification and extensions for additional information. Its specification index distinguishes versions and extension families. Application and extension support must be checked; a 3MF file is not universally printer-ready machine instructions. If using STL, record the intended unit interpretation explicitly. [C02]

This reference stops at the asset handoff. The companion **Blackwish 3D Printing Research Reference** owns fabrication processes, materials, slicing, machine-specific constraints and physical qualification. Neither a successful render, mesh validator nor slicer preview proves a physical object safe for a functional use. No print, purchase or machine action is authorized here.

Original counterexample: a miniature museum bench looks correct and has a closed surface, but its thin supports are unsuitable for the chosen process. The artistic contract can pass while the fabrication contract fails. Route it for separate process-specific review instead of weakening the acceptance language.

A proposed Blackwish asset record

This is a conceptual schema for discussion, not deployed code, a canonical amendment or a validated instrument. Distinguish claims, evidence, transformations and decisions; avoid one universal "safe" flag.

Identity and intended use

asset_id identifies the conceptual asset; **version_id** identifies a frozen revision; **representation_id** identifies a master, preview, interchange file or delivery derivative. Record family, plain-language description, owner or steward, intended audience, use, target platform and locale. Keep identity stable when a filename changes, but assign a new version when relevant content changes.

Source and dependency graph

For each input, record source type, creator or provider as known, acquisition date, source reference, input digest and dependency relationship. Textures depend on material sources; dubbed lines depend on scripts and recordings. Capture unknown origins explicitly. Preserve originals and store credentials or sensitive consent evidence outside broadly distributed asset packages.

Transformation recipe

Record tools, versions, relevant plug-ins, parameters, model or checkpoint identity where available, random seed if used, prompts and conditioning inputs where retention is appropriate, and any human edits. State reproducibility: byte-identical in a specified environment; tolerance-equivalent under stated bounds; or traceable but not reproducible, with the output preserved. A seed alone is not a guarantee across versions, devices, execution paths or hosted-service changes.

Technical and semantic contract

Record units, axes, pivot, dimensions, coordinate space, frame or sample rate, color interpretation, channel meanings, language and dependencies as applicable. Reference a target profile rather than duplicating every platform rule in each record. Use explicit not-applicable values where fields genuinely do not apply. Do not apply mesh checks to text.

Rights and use scope

Store the actual license or agreement reference, version or date, attribution requirements, modification and redistribution constraints, model or likeness consent references where relevant, and unresolved questions. SPDX identifiers can reduce ambiguity for recognized licenses, but an identifier does not decide whether the evidence covers this asset and use. Ownership, license and consent are separate questions. [C01]

Evidence and authority

Each test entry needs subject version, method, environment, result, timestamp, artifact or log reference, and reviewer if applicable. Each human decision needs action, subject package, scope, decision maker and date. Keep text ratification separate from deployment authorization. Release status should be derived from required evidence and authority, not editable by a generator announcing success.

Minimal viable record: identity, intended use, master location, dependencies, origin and rights evidence, transformation summary, target checks, unresolved issues and explicit release decision. Expand only for a real retrieval, debugging or governance need.

Validation is a set of claims and tests

A validator is useful when its test addresses the relevant failure. Format checks do not prove artistic adequacy or universal interoperability. Select from these proposed layers according to the asset contract.

Layer	Example evidence	What it does not prove
Structural	Schema, parser, dependency and checksum checks	Meaning, quality or lawful use
Technical	Import, playback, render and target profiling	Fit for every platform
Perceptual	Review at actual size and intended conditions	General audience acceptance
Semantic	Text, continuity, labels and branch checks	Every possible interpretation
Rights	License and consent review for the named use	Technical compatibility
Authority	Explicit decision covering the package and action	Automatic permission for later revisions

Use negative tests as well as happy paths

Hypothetical counterexamples reveal the boundaries of a rule. Remove a texture; substitute a missing font; change units; shorten a clip; use an unsupported extension; introduce a right-to-left locale; reference an unavailable audio file; invalidate one dependency. Would the process notice and contain the failure? These are paper or local-artifact design checks, not experiments on people or changes to live systems.

For geometry, review topology, normals, scale, collisions, deformations and target display. For images, review alpha edges, color handling, readability and compression. For audio, review clipping, noise, loops, channel routing, loudness for the destination and intelligibility. For narrative, inspect unreachable states, missing endings, references, localization and consistency. A checker may cover one layer while the others remain unknown.

Define what counts as enough evidence

A screenshot does not validate a full animation, a sample import does not validate every asset, and a master pass does not validate its derivative. Record the population, sample strategy, checks performed and known omissions. When a test is sampled, report it as sampled, not complete.

Tie regressions to change

Use a small reference set that exercises known difficult cases: transparency, unusual characters, deformation extremes, multi-channel audio, material variants and non-default scale. A tool upgrade or exporter change should trigger comparison against that set before broad conversion. Preserve both outputs and the known-good package.

Exit condition: the asset contract's required tests pass on the actual candidate, remaining limitations are explicit and accepted by the appropriate reviewer, rights evidence fits the use, and any release action has separate authorization. Otherwise, name the precise blocker rather than granting ceremonial approval.

Compare total usable cost rather than generation speed

A cheap candidate can be expensive to ship. A costly handmade master can be economical when it supports many controlled revisions. No current market price, measured Blackwish throughput or guaranteed savings estimate is asserted here.

A practical cost model

Compare **acquisition or creation + cleanup + adaptation + validation + rights review + integration + storage and compute + revision and maintenance + expected rework**. Include staff attention and switching costs. The categories can be estimated in hours before assigning monetary rates. Use ranges and identify which uncertainty dominates rather than disguising a guess as a precise total.

Make when identity, editability, unusual constraints or long-term control dominate. **License** when a suitable asset already exists and the rights and adaptation burden are clear. **Commission** when specialist judgment is required but an internal capability is not justified. **Proceduralize** when repeated variation follows stable rules. **Use generative candidates** when rapid exploration is valuable and reconstruction, review and rights questions are included in the plan.

Three transparent calculations

A 2048 by 2048 RGBA8 image contains 16 MiB of uncompressed pixel data at its largest level. A complete conventional two-dimensional mip chain adds approximately one third, bringing the total to about 21.33 MiB. This is an arithmetic illustration, not a GPU-residency prediction: compression, format, padding, streaming and engine behavior alter actual use.

One minute of stereo PCM at 48,000 samples per second and 24 bits per sample contains 17,280,000 bytes of sample data, before a container's overhead. Compressed delivery can be much smaller, but the editable recording should not be discarded solely because its delivery copy is compact.

At 60, 90 and 120 frames per second, total frame time is about 16.67, 11.11 and 8.33 milliseconds respectively. That is the whole frame's budget, not an entitlement for any one asset. CPU work, GPU work, synchronization and target-device behavior require actual profiling.

Track usable yield

Record how many candidates survive each gate and why others fail. A system producing 100 images with two usable results should not be compared with a system producing ten images with eight usable results using output count alone. Define "usable" before counting; the denominator includes rejected and abandoned work.

Decision rule: prefer the method whose credible total cost and controllability fit the contract. Stop generating when the missing ingredient is a decision, a rights record, a target test or an editable structure. More candidates cannot resolve those deficits by volume alone.

Three original end to end examples

These hypothetical examples are G-rated in intent, nonfantasy and untested. They show how the same framework changes with the medium; they are not production estimates or instructions to deploy.

A community museum information sign

Contract: a readable in-scene sign, a print illustration and localized accessible text. Author the lettering as real text or controlled vector paths, retain an editable layout and create a simple dimensioned 3D support. Keep the sign's informational content as a separate structured record rather than treating the painted texture as the source of truth.

Derivatives: export a runtime mesh with the agreed pivot and collision, a texture or material suitable for the selected engine, a print image at the needed physical size, and a text description. Review the runtime sign from the intended camera distance and the print image at actual scale. The localized edition may need a different layout, not just replacement words.

Counterexample: an AI image looks excellent but contains incorrect lettering and a stand with impossible geometry. Keep it as a visual concept if appropriate; do not relabel it an editable, technically valid production sign. Stop if the museum's factual claims have not been verified or the required typeface rights are unknown.

A park maintenance worker character

Contract: an original adult character who walks, points and lifts an empty watering can in an educational scene. Build a reference sheet, clear silhouette and clothing that supports the required movements. Retain the mesh, rig, weights and source clips. Use motion capture only with appropriate permission and account for cleanup and retargeting.

Derivatives: produce the selected runtime skeleton and clips, a still illustration and a captioned short sequence. Review shoulders and wrists under the actual gestures, hand contact with the handle, foot sliding, collision and transitions. A still portrait does not certify the rig. A neutral test pose does not prove the lifting motion works.

Counterexample: the same gesture reads as a wave when the camera changes. Semantic intent needs review in context, not merely a motion-quality score. Do not infer consent to reuse a performer's likeness or voice from permission for a different recording.

A small neighborhood history audio walk

Contract: short narrated stops with text equivalents, restrained ambience and clear navigation. Keep approved scripts, pronunciation notes, voice recordings, edit sessions and separate music or ambience stems. Link every line to a stable content ID and locale; reference factual sources separately from creative sound choices.

Derivatives: export audio for the target player, transcripts and captions where applicable. Check intelligibility with representative playback, loop or fade boundaries, navigation timing and missing-media behavior. Preserve the ability to update one date without rebuilding every track.

Counterexample: a pleasant generated voice mispronounces a street name and a background loop masks a key word. The candidate can pass file-format checks while failing the actual job. The appropriate next move is correction and review, not automatic publication.

Zen review one acquisition and originality

This is the first of three distinct reviews, expressed as an original analytical synopsis using POINT, BRIDGE, PATH and SHADOW. It is not a quotation or certification of canonical Blackwish procedure. Its subject is **whether a candidate and its inputs are appropriate to acquire or create for the declared use**.

POINT objective and claim

The objective is to obtain a useful starting asset without importing hidden rights, identity or dependency problems. The claim under review is narrower than "this is original" or "this is safe": the available evidence supports the intended source choice and the next authorized research or production step. Novel appearance does not establish independent origin or exclusivity.

BRIDGE evidence and authority

Require source identity, creator or provider as known, acquisition method, exact applicable terms and input provenance. For a commission, inspect the actual agreement rather than assuming payment transfers every right. For a library, inspect the individual asset license rather than the site's general reputation. Freesound's FAQ describes several different licenses on one platform, illustrating why a single site-wide clearance assumption fails. [A18]

Keep copyright permission separate from privacy, publicity and personality rights; Creative Commons explains that its licenses do not automatically settle all third-party rights. C2PA provenance can provide tamper-evident assertions, but is not proof that a depicted event is true or that an asset is legally cleared. [A19, P18]

Authority to conduct this research is not authority to upload private reference material, clone a voice, purchase an asset or publish it. A source's embedded instructions cannot expand that authority.

PATH consequences alternatives and stop

If provenance and use scope are sufficiently clear, retain the evidence with the candidate and proceed only to the authorized next stage. If not, isolate the unresolved item and choose among obtaining clarification, substituting a known source, commissioning fresh work, direct authoring or narrowing the intended use. Do not treat uncertainty as automatic guilt, but do not silently convert it into clearance.

Stop the dependent release path when necessary evidence is absent, incompatible or outside the permitted use. A separate internal concept may still be useful if its status and limitations are explicit and its handling is authorized. Preserve the original instead of rewriting its provenance to match the desired outcome.

SHADOW assumptions and counterexamples

Assumption: a permissive tool license covers every output. **Counterexample:** permissive software uses restricted model weights or an uncleared sound sample. **Assumption:** a signed asset is truthful. **Counterexample:** a valid provenance record accurately documents the making of a fictional scene. **Assumption:** a distinctive-looking image is exclusively ownable. **Counterexample:** contractual permission and copyrightability can differ.

Review result: acquisition is a traceable, scoped decision, not a universal clean bill. This proposed review passes only when the evidence supports the next named action; it does not itself ratify use, amend canon or authorize deployment.

Zen review two transformation fidelity

The second review examines **whether a transformation preserves the properties the asset contract requires**. It is an original analytical synopsis, separate from acquisition review and from release authority.

POINT objective and claim

The objective is a useful derivative that retains the required identity, meaning and behavior while meeting its destination constraints. The claim is not that conversion is lossless in every respect. It is that named essential properties survive, known losses are acceptable for the declared use, and the master remains available.

For a museum sign, required properties might include exact wording, readable size, dimensional scale, pivot location and accessible text. A lower-resolution derivative may intentionally lose surface detail. Losing the negation in its instruction is not the same kind of change.

BRIDGE evidence and authority

Compare the actual input and output revisions using the relevant evidence: dimensions, fixed-camera views, target import, deformation suite, timing checks, channel interpretation, dependency inventory and semantic review. A format validator addresses structure; it cannot settle the sign's truth or the character's gesture meaning. A screenshot establishes only what that view shows.

Record the toolchain and settings. A random seed is insufficient to establish cross-version identity; preserve the actual output and define the reproducibility claim. Keep authoring controls separate from exported data. A skeleton transfer does not imply that a custom control rig remains editable, and a flattened mix does not imply that its stems are recoverable.

The authority to compare two local candidate files does not authorize replacing a released asset. A reviewer may accept a technical deviation only within the scope they actually control.

PATH consequences alternatives and stop

If required properties survive, attach the evidence to the derivative and retain known limitations. If they fail, choose repair, a different export profile, a richer interchange format, a simpler target feature set or a different source. It can be rational to retain two delivery variants instead of forcing one file to satisfy incompatible targets.

Stop when a required property is lost and cannot be recovered or explicitly revised by the appropriate decision maker. Do not discard the source to make the failure disappear. Do not reclassify an inaccessible or noneditable output as acceptable merely because the visual result is compelling.

SHADOW assumptions and counterexamples

Assumption: opening the file proves portability. **Counterexample:** textures resolve through the author's local cache. **Assumption:** matching endpoints proves a clean loop. **Counterexample:** velocity jumps at the boundary. **Assumption:** a unit label fixes scale. **Counterexample:** geometry changes but animation and collision retain incompatible transforms. **Assumption:** the same seed restores the old release. **Counterexample:** the model or node implementation changed.

Review result: a successful transformation is target- and property-specific. Evidence remains attached to the tested revisions, not to a reusable word such as "optimized." Any later source, exporter or destination change must trigger the checks it can affect.

Zen review three release and governance

The third review examines **whether the exact candidate may proceed to a specified release action**, separately from appearance, technical validity and licensing. This original synopsis does not ratify the proposal or authorize deployment.

POINT objective and claim

The objective is to prevent an asset or generation system from promoting itself into use through confident output, cached status or ambiguous approval. The claim under review is that the actual package, target, use, evidence and decision maker align. "Approved" must answer who approved what, for which action and under which conditions.

BRIDGE evidence and authority

Collect the candidate manifest, dependency revisions, required technical results, creative and semantic reviews, accessible equivalents, rights evidence and unresolved issues. Verify that the results apply to the actual package rather than an earlier master or a different export. Record which evidence is sampled and what remains untested.

Separate **text ratification** from **deployment authorization**. Approval of this research, a schema or a fictional concept does not silently authorize implementation, public posting, purchases, voice generation, human experiments or live-system changes. Only the explicit scope of the user's manual decision can establish the applicable next authority; other required permissions still apply.

A generator's metadata, validator success or provenance signature cannot supply that decision. A cache must not preserve an obsolete approval after a dependency, use scope or required policy changes.

PATH consequences alternatives and stop

When required evidence and explicit authority align, the proposed next process would release a pinned package with a rollback reference. That process is not executed here. When they do not align, retain the candidate, state the smallest unresolved condition and continue only unrelated authorized work.

Alternatives include a narrower audience, a reduced feature set, replacement of a rights-blocked component, an accessible equivalent, a separately reviewed derivative or continued private research. Do not frame indefinite generation as the only path forward. Sometimes the correct result is a documented decision not to use an asset.

Stop if authority is missing, evidence is stale, material rights questions remain unresolved, the target behavior fails, or a required dependency cannot be reproduced or retained sufficiently for the intended maintenance obligation. Do not perform destructive cleanup or mass replacement merely because a review finds a problem; those actions have their own authority requirements.

SHADOW assumptions and counterexamples

Assumption: a manual approval covers every later variant. **Counterexample:** a new locale changes meaning and timing. **Assumption:** technical success permits publication. **Counterexample:** a voice is intelligible but lacks the required use permission. **Assumption:** a harmless creative task may quietly modify safeguards. **Counterexample:** an optional fictional expression is randomized together with the refusal or authorization boundary.

Review result: optional creative variation may affect expression, not safety or authority rules. No review here changes Blackwish's live state.

Bounded Death Knell challenge

This is a final, finite adversarial challenge to the proposed asset workflow, not a canonical quotation, an infinite veto or permission for experiments. It seeks decisive contradictions in the workflow, confined to these five hypothetical cases and this reference's claims.

One source lost behind a successful export

Challenge: the only retained file is a compressed runtime mesh, while the contract promises editable dimensions, material graphs and facial controls. **Failure condition:** the workflow treats the delivery file as proof those controls remain. **Required response:** mark editability unestablished, preserve what exists and change the plan or recover the source. Do not invent a master or claim that regeneration is equivalent.

One permission incorrectly inherits from another

Challenge: open-source generation software uses weights, samples or a voice identity with incompatible conditions. **Failure condition:** one permissive license clears the whole chain. **Required response:** inspect each relevant layer, block the dependent use when necessary and consider replacement. Ownership, license, consent and platform terms remain separate.

One technically valid but semantically wrong result

Challenge: a sign's text loses "not," a caption reveals a hidden identity, or a navigation layout has disconnected destinations. **Failure condition:** parser success or visual quality overrides meaning. **Required response:** semantic and context checks fail independently; repair the actual claim and invalidate affected derivatives.

One supposedly stable regeneration changes the asset

Challenge: the same seed is rerun after a model, node, library or platform change. **Failure condition:** the new output silently replaces the old approved version. **Required response:** preserve the approved binary, create a distinct candidate, record changed dependencies and compare the properties that matter. Reproducibility is scoped evidence, not a slogan.

One review silently becomes authority

Challenge: a positive technical review triggers publication, purchases, new recording or changes to safeguards. **Failure condition:** the process derives authority from its own output. **Required response:** stop the dependent action and require the applicable explicit manual authorization. Research and hypothetical analysis can continue within their existing scope; live changes cannot inherit permission from that research.

Conditions for ending this challenge

The bounded review can conclude when each case has a clear disposition, no unresolved contradiction undermines the claimed workflow, limitations are recorded and no actual action has been smuggled into the analysis. A pass means only that these five desk challenges did not defeat the stated proposal. It is not proof against every failure, rights clearance, user testing, certification or deployment approval.

Current disposition: the proposed separation of masters, derivatives, evidence, rights and authority addresses these cases at the design level. Operational performance remains unestablished because no Blackwish implementation or asset batch was run. The next meaningful evidence would be a separately authorized, narrowly specified production check, not more rhetorical approval.

Practical conclusions and next decisions

The strongest general strategy is a mixed pipeline: direct authorship for defining choices, procedural recipes for controlled families, capture for observed detail, learned generation for selected candidate work, and explicit review for everything that must be true. The mix should follow the job, not a commitment to make every asset by the same method.

Decide these before selecting a tool stack

1. **Primary destination and audience.** A small interactive scene, illustrated book and physical object require different evidence.
2. **First asset family.** Choose a representative family with a real revision need, not the most visually impressive demo.
3. **Required editability.** Name the properties that must remain under direct control.
4. **Rights and data constraints.** Identify what may be uploaded, acquired, generated, retained and distributed.
5. **Acceptance and authority.** Define tests, reviewers, unknown states and the specific action requiring manual ratification.

A useful future comparison could use one ordinary asset family, such as museum signs, made through direct authoring, an approved template and a generated-candidate route. The question would be which yields maintainable, rights-supported, target-ready results at acceptable total effort. Such a comparison is only a proposal here; no generation experiment, human study or live deployment was performed.

Preserve a small set of durable distinctions

Asset versus file: one conceptual object can have many representations. **Master versus derivative:** richer editability and destination efficiency serve different purposes. **Provenance versus truth:** a reliable history can document fictional content. **License versus ownership versus consent:** permission has multiple independent bases. **Generated versus validated:** production of output supplies no automatic acceptance evidence. **Ratified text versus deployed system:** a review decision must not silently become an action.

Deterministic versus traceable: exact regeneration is a stronger claim than preserving the output and its recipe. **Local rule versus global property:** matching tiles do not prove a connected route. **Structural versus semantic validity:** a well-formed file can convey the wrong information. **Rendering versus manufacturing:** a visual surface can be unsuitable for fabrication or functional use.

What remains uncertain

The actual Blackwish engine, production hardware, distribution territories, budget, final audience, installed tool versions and approved asset contract are not established in this research. Accordingly, this reference does not rank vendors by measured output quality, prescribe universal performance budgets, quote current prices or clear rights to a specific asset. Rolling documentation and licenses must be rechecked at the point of adoption and release.

Recommended next decision: select one destination and one asset family, then agree on its contract and evidence requirements. Keep the first scope small enough that a failure teaches something specific and a successful result remains editable. No current proposal changes canon, optional fictional mechanics, safeguards or deployment status.

Source access ledger 1

46 unique primary source URLs are cited in this reference. All were checked on 23 October 2026. Opened means the relevant retrieved page or passage was inspected, not that an entire manual, repository or specification was read. Search extract identifies a narrower access level. Linked titles open the exact cited source.

V01 Khronos, glTF registry
Khronos | Opened relevant primary page. current registry identifies glTF 2.0. Runtime delivery purpose and specification registry

V02 Khronos, glTF specification
Khronos | Opened relevant primary page. glTF specification 2.0.1, front-matter date 2021-10-11. Coordinates, units, material/texture semantics

V03 Pixar/OpenUSD, linear units
Pixar/OpenUSD | Opened relevant primary page. release documentation. Stage-wide units, centimeter fallback, assembly correction

V05 Pixar/OpenUSD, instancing
Pixar/OpenUSD | Opened relevant primary page. release documentation. Shared scenegraphs, edit restrictions, unstable prototype paths

V06 MaterialX/ASWF, specifications
MaterialX/ASWF | Opened relevant primary page. page labels current specification v1.39. Core, standard, physical, and NPR node specification families

V07 W3C, SVG 2
W3C | Opened relevant primary page. Candidate Recommendation Snapshot 2026-10-06. Vector/mixed content and dynamic document capability

V08 W3C, PNG
W3C | Opened relevant primary page. Recommendation 2025-06-24. Lossless raster, alpha and color metadata

V10 OpenEXR/ASWF
OpenEXR/ASWF | Opened relevant primary page. latest documentation. Premultiplied convention, channels, data/display windows

V11 Khronos, glTF Validator
Khronos | Opened relevant primary page. repository. Official structural validation tool

V12 Khronos, KTX
Khronos | Opened relevant primary page. KTX 2.0 discussion. GPU texture container, mipmaps, supercompression/transcoding

V13 Mildenhall et al., NeRF
Mildenhall et al. | Opened relevant primary page. ECCV 2020 research. Radiance-field representation and novel-view synthesis

V14 Kerbl et al., Inria et al.
Kerbl et al. | Opened relevant primary page. SIGGRAPH 2023 research. Gaussian representation and benchmark-limited performance claims

Source access ledger 2

All entries checked 23 October 2026. Access describes retrieved evidence, not exhaustive reading or software testing. Source titles link to the exact URLs. Rolling documentation and terms should be rechecked before adoption.

V16 COLMAP project
COLMAP project | Opened relevant primary page. current tutorial. SfM/MVS reconstruction pipeline and capture conditions

V18 Pixar/OpenUSD, UsdSkel
Pixar/OpenUSD | Opened relevant primary page. release documentation. Skeleton/binding/animation separation and blendshapes

V19 W3C WAI, non-text contrast
W3C WAI | Opened relevant primary page. WCAG 2.2 explanatory document. Applicable 3:1 threshold, exceptions, meaningful graphics

V20 Microsoft, OpenType overview
Microsoft | Opened relevant primary page. OpenType 1.9.1 overview. Glyph substitution, positioning, variation and multilingual shaping

A01 Musical Compositions and Sound Recordings
U.S. Copyright Office | Opened. undated page. Composition and sound recording are separate works.

A02 Setting Source and Destination Properties
Audiokinetic | Search extract; direct open attempted 2026-10-23 returned HTTP 403. rolling edge help. Source/destination cues, transition fades, post-exit behavior.

A04 Professional Voice Cloning
ElevenLabs | Opened. rolling documentation. Own-voice verification for Professional Voice Cloning; recording contamination and cross-language pronunciation caveats.

A05 Importing Audio Samples
Godot Engine | Opened. stable documentation. WAV/Ogg Vorbis/MP3 imports, compression and loop considerations.

A06 EBU R 128 Loudness Recommendation
European Broadcasting Union | Opened. R 128 version 5.0, November 2023. Programme loudness recommendation of -23 LUFS and loudness/true-peak descriptors.

A08 Ink Interactive Narrative Repository
inkle | Opened. rolling official repository. Interactive narrative language and division of editor/compiler/runtime/integrations.

A10 Localization and Assets for Godot GDScript
Yarn Spinner | Opened. Godot GDScript documentation, page navigation identifies v3.2 current. Line IDs link translation, voice and other assets; system does not itself translate.

A11 ICU Formatting Messages
Unicode/ICU | Opened. current user guide, legacy MessageFormat section. Locale-aware plural selection and structured message formatting.

Source access ledger 3

All entries checked 23 October 2026. Access describes retrieved evidence, not exhaustive reading or software testing. Source titles link to the exact URLs. Rolling documentation and terms should be rechecked before adoption.

A13 Captions and Subtitles
W3C WAI | Opened. current media guide. Captions represent speech and important non-speech audio.

A14 Audio Description of Visual Information
W3C WAI | Opened. current media guide. Description conveys visual information; advance planning helps production.

A15 Xbox Accessibility Guideline 104
Microsoft | Opened. Xbox Accessibility Guideline 104. Subtitle/caption coverage, speaker/directional information and configuration.

A17 Complex Images
W3C WAI | Opened. Images Tutorial, Complex Images. Short identification plus fuller representation for complex diagrams/charts.

A18 Freesound Frequently Asked Questions
Freesound / Universitat Pompeu Fabra | Opened. No release pin. Per-sound CC0, BY, BY-NC licensing and attribution distinctions.

A19 Creative Commons Frequently Asked Questions
Creative Commons | Opened. official FAQ. Copyright licenses do not automatically clear third-party personality/privacy/publicity rights.

P02 SideFX, Introduction to PDG and TOPs
SideFX | Opened relevant primary page. Dependency work items and selective recomputation. Houdini 22.0 page header; architecture application is original synthesis.

P03 Epic Games, Procedural Content Generation Overview
Epic Games | Opened relevant primary page. Graph based spatial content generation and point attributes. Rolling documentation; not an engine licensing review.

P04 Google OR-Tools, CP-SAT README
Google OR-Tools | Opened relevant primary page. Constraint programming and feasibility machinery. Moving stable branch; no performance comparison.

P05 Maxim Gumin, WaveFunctionCollapse
Maxim Gumin | Opened relevant primary page. Example driven bitmap and tile generation. Moving repository; global connectivity caution is original analysis.

P08 Microsoft, TRELLIS
Microsoft | Opened relevant primary page. Structured 3D generation and text versus image routes. README update 2025-03-25; not a comparative benchmark.

P09 Tencent, Hunyuan3D-2 LICENSE
Tencent | Opened relevant primary page. Hunyuan3D-2 model license, release 2025-01-21. Territorial exclusions and commercial conditions; not generalized to later models.

Source access ledger 4

All entries checked 23 October 2026. Access describes retrieved evidence, not exhaustive reading or software testing. Source titles link to the exact URLs. Rolling documentation and terms should be rechecked before adoption.

P12 Meta, AudioCraft
Meta | Opened relevant primary page. Audio generation repository and separate MIT code and CC-BY-NC-4.0 pretrained weight licenses.

P16 U.S. Copyright Office Copyright and Artificial Intelligence Part 2
U.S. Copyright Office | Opened relevant primary page. Human authorship and AI outputs in United States copyrightability analysis. Published 2025-01-29; not a judicial ruling or global rights clearance.

P17 OWASP, LLM Prompt Injection Prevention Cheat Sheet
OWASP | Opened relevant primary page. Untrusted content and permission enforcement at tool boundaries. Specific quarantine design is a proposal.

P18 C2PA, Content Credentials Explainer v2.2
C2PA | Opened relevant primary page. Provenance assertions versus truth and completeness. Inspected explainer v2.2; not claimed latest.

P19 PyTorch, Reproducibility
PyTorch | Opened relevant primary page. No guarantee across releases, platforms or CPU/GPU with identical seeds. Versioned 2.14 documentation; updated 2026-05-14.

P20 OpenUSD, Introduction to USD
OpenUSD | Opened relevant primary page. Scene composition and interchange; not a rigging system. Rolling page returned 26.11 label; not an installed or released compatibility claim.

C01 SPDX License List
SPDX | Opened relevant portions. 3.29.0 dated 2026-09-16. Standard license identifiers, license names and canonical license text; not an assessment of this project's legal rights.

C02 3MF Specifications
3MF Consortium | Opened specification index. Listing distinguishes ISO core 1.3.0 and latest-revision core 1.4.0. 3MF manufacturing purpose and extension families; not full conformance review or proof of printability.

C03 Timeline Structure
OpenTimelineIO | Opened relevant documentation. 0.19.0.dev1 documentation at access. Timeline, tracks, clips and media references; effects and unavailable media ranges require downstream handling.

C04 OpenColorIO overview
OpenColorIO | Opened overview. Homepage announces 2.5.0 at access. Color-management role and application integration; not verification of a project's particular config or calibration.